Business Data Management Capstone Project Final Submission

Project Title – Maximizing Revenue and Improving Inventory Management in a Bakery: A Data Driven Approach

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1. Executive Summary:

This project began with a simple yet ambitious goal: to help Golden Crust Bakery in Kanpur maximize revenue and improve inventory management through data-driven insights. From the initial proposal to the final submission, the journey has been one of discovery, analysis, and actionable transformation.

In the proposal phase, the focus was on understanding the bakery's challenges—inefficient inventory management and a lack of sales tracking. The plan was to collect 30-40 days of sales and inventory data manually, as the bakery had no digital records. The proposal outlined the use of Excel for analysis, with tools like pivot tables, charts, and ABC classification to uncover trends and patterns.

By the mid-term submission, significant progress had been made. Over 30 days, 2,523 sales entries were meticulously recorded, and preliminary analysis revealed key insights. Rich Cream products emerged as the top revenue generators, while Snacks and Bread items lagged. Temporal trends showed weekends as the busiest periods, with Saturdays generating ₹41,145 compared to Tuesdays' ₹1,750. ABC classification was introduced, categorizing products into high-priority (Class A), moderate (Class B), and low-priority (Class C) items. This laid the foundation for targeted inventory management.

In the final submission, the analysis deepened, and actionable strategies were developed. High-sales but low-margin products like R01 were identified, prompting recommendations for cost optimization and supplier renegotiations. Bundling strategies were proposed to boost slow-moving items, and staffing adjustments were suggested to meet weekend demand. The project culminated in a comprehensive plan to introduce "Weekend Specials" and revise production schedules, projected to increase annual profitability by 15–20%.

This journey from proposal to final submission has not only provided Golden Crust with actionable insights but also demonstrated the power of data in transforming small businesses. By bridging analytics with practical solutions, the bakery is now equipped to achieve sustainable growth and operational efficiency.

2. Detailed Explanation of Analysis Process/Method:

2.1 Data Collection:

- Data was manually collected over 30 days, resulting in 2,523 sales entries.
- Each entry included detailed information such as product name, type, SKU, cost price, selling price, opening stock, closing stock, units sold, date, and day of the week.
- Data was recorded daily in Excel to ensure accuracy, as the bakery had no prior digital records. This involved close collaboration with the bakery owner and staff to maintain consistency.

2.2 Data Cleaning:

- The dataset was thoroughly checked for missing values, duplicates, and inconsistencies. Fortunately, none were found due to the meticulous manual recording process.
- Columns like "Cost Price" and "Selling Price" were formatted as currency for clarity and ease of analysis.
- Two new columns, "Sales" and "Profit," were added to enhance financial analysis. These were calculated as:
    i. Sales = Selling Price × Units Sold
    ii. Profit = (Selling Price − Cost Price) × Units Sold
- This step ensured the dataset was clean, structured, and ready for deeper analysis.

2.3 Data Preprocessing:

- Since the data was manually recorded in a structured format, minimal preprocessing was required.
- No categorical encoding or numerical scaling was needed, as the dataset was already in a usable format.
- the primary transformation involved organizing the data by product type, SKU, and day of the week to enable granular analysis. For example, sales data was grouped by weekdays and weekends to identify temporal trends.
- Additional calculations, such as cumulative weekly sales and average sales per product category, were performed to support the analysis. These calculations helped identify trends over time and provided a clearer picture of product performance across different categories.

2.4 Analysis Methodology:

- Tools Used: **Excel** was the primary tool, leveraging features like pivot tables, filtering, and charting tools for analysis. ABC analysis was also applied to classify products based on their sales contribution.

2.4.1 Steps:

- Descriptive Analysis: Calculated key metrics such as average daily sales, median sales, and standard deviation to understand overall sales performance.
- ABC Classification: Products were categorized into three groups based on their contribution to total sales:
- Class A: Top 70% of sales (high-priority items).
- Class B: Next 20% of sales (moderate-priority items).
- Class C: Bottom 10% of sales (low-priority items).
- Temporal Analysis: Sales trends were analyzed across days of the week, with a focus on identifying peak and off-peak periods. For example, Tuesdays were identified as the bakery's off day, with significantly lower sales compared to weekends.
- Profitability Analysis: Compared sales and profit margins across product categories to identify high-revenue but low-margin items.

2.5 Interpretation and Validation:

- Findings were cross-verified with the bakery owner to ensure alignment with real-world observations. For example, the high weekend sales trend matched the owner's experience of increased customer footfall on Saturdays and Sundays.
- Visualizations like bar charts and line graphs were used to present trends clearly, making it easier for the owner to understand and validate the insights.
- The iterative process of analysis and validation ensured that the results were both accurate and actionable.

3. Results and Findings:

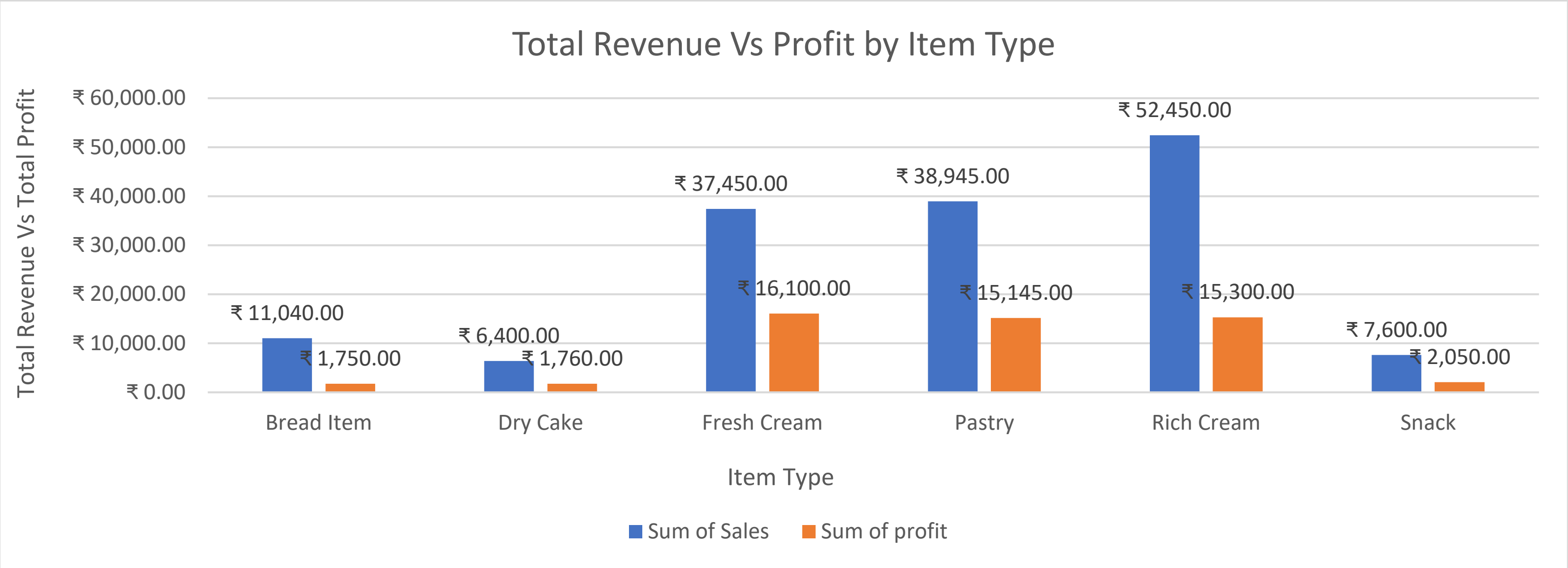

Figure 1 (Total Revenue Vs Profit by Item Type)

3.1 Overview of Revenue and Profit Trends:

3.1(a) Rich Cream Generates the Highest Revenue:

- Rich Cream has the highest revenue at ₹52,450 but a moderate profit of ₹15,300.
- This suggests that while it is a high-selling item, its profit margin is not significantly higher compared to other items.

3.1(b) Fresh Cream Has the Best Profit Margin:

- Fresh Cream records the best profit margin with ₹16,100 profit on ₹37,450 revenue.
- Despite having lower revenue than Rich Cream, it provides better profitability, indicating a higher return on sales.

3.1(c) Pastry is a Strong Performer in Both Revenue and Profit:

- Pastry generates ₹38,945 in revenue with a ₹15,145 profit.
- This makes it a well-balanced product with a strong combination of both sales volume and profitability.

3.1(d) Low Revenue and Profit for Bread and Dry Cake:

- Bread Item and Dry Cake have significantly lower revenues of ₹11,040 and ₹6,400, respectively, with very minimal profits (₹1,750 and ₹1,760).

- This suggests that these products might not be major revenue drivers and could be reconsidered in terms of pricing or promotional strategies.

3.1(e) Snack Items Have a Better Profit-to-Revenue Ratio:

- Snacks generate ₹7,600 in revenue with a ₹2,050 profit.
- While the revenue is lower, the profit ratio compared to revenue is better than Bread and Dry Cake.

3.1(f) Recommendations:

- Focus on Fresh Cream and Pastry: They offer good profitability and stable revenue, making them ideal for maximizing earnings.
- Re-evaluate Bread and Dry Cake: Either improve pricing, marketing, or discontinue if they are not contributing much to profits.
- Optimize Rich Cream's Cost Structure: Since it has the highest revenue but moderate profit, consider cost reduction or price adjustments to improve its profitability.

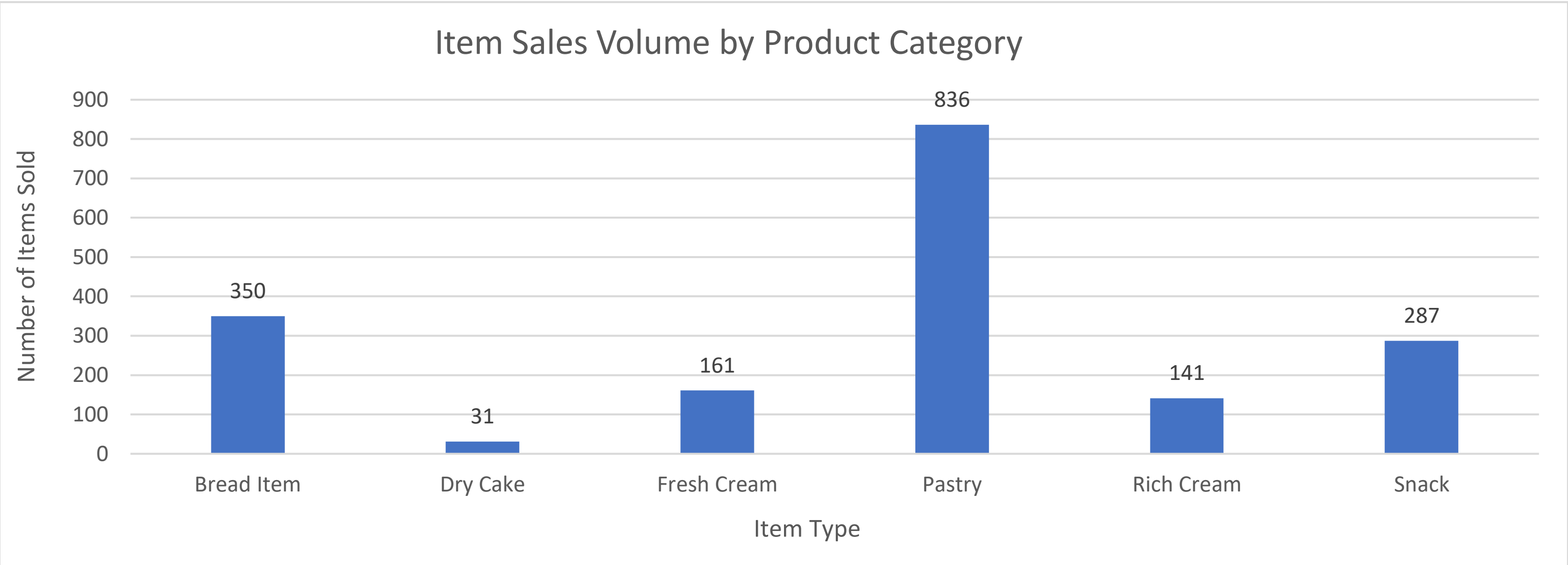

Figure 2 (Item Sales Volume by Product Category)

3.2 Analysis of Product Categories

3.2(a) Bread Items – High Sales, Low Revenue & Profit

- Observation: Bread items recorded a high sales volume (350 units) but contributed only ₹11,040 in revenue and ₹1,750 in profit.
- Insight: The high sales volume indicates strong demand, but low revenue and profit margins suggest pricing issues or high production costs.

- Recommendation:
    - Adjust pricing strategies by introducing premium bread variants (e.g., multigrain, sourdough) to increase margins.
    - Implement value packs or bundle offers to encourage bulk purchases.

3.2(b) Dry Cake – Lowest Performing Category

- Observation: Dry cakes had the lowest sales (31 units), generating ₹6,400 in revenue and ₹1,760 in profit.
- Insight: The minimal demand suggests poor customer preference or insufficient marketing efforts.
- Recommendation:
    - Consider discontinuing this item if it remains unprofitable.
    - Revamp the product line with new flavours, improved packaging, or promotional discounts to boost sales.

3.2(c) Fresh Cream – Moderate Sales with Good Revenue

- Observation: Fresh cream products sold 161 units, generating ₹37,450 in revenue and ₹16,100 in profit.
- Insight: This category demonstrates strong revenue generation, indicating customers are willing to pay a premium.
- Recommendation:
    - Introduce new flavours and premium variations to expand customer choices.
    - Offer bundle deals to increase average purchase value.

3.2(d) Pastries – Best-Selling Category with Strong Profitability

- Observation: Pastries had the highest sales volume (836 units), generating ₹38,945 in revenue and ₹15,145 in profit.
- Insight: Pastries are a key revenue driver and highly preferred by customers.

- Recommendation:
  - Expand pastry offerings with seasonal or festive editions.
  - Ensure sufficient inventory and production to meet demand.

3.2(e) Rich Cream – Highest Revenue but Moderate Profit

- Observation: Rich cream products generated the highest revenue (₹52,450) but yielded only ₹15,300 in profit despite selling just 141 units.
- Insight: The high revenue suggests strong demand, but profitability is constrained, possibly due to high production costs.
- Recommendation:
  - Optimize ingredient sourcing and reduce production costs.
  - Introduce cost-effective packaging to improve profit margins.
  - Offer limited-time discounts on bulk purchases to encourage higher sales.

3.2(f) Snacks – Moderate Sales with Low Profit Margins

- Observation: Snacks sold 287 units, generating ₹7,600 in revenue and ₹2,050 in profit.
- Insight: While sales volume is decent, the profit margin is relatively low.
- Recommendation:
  - Introduce high-margin snack variants.
  - Offer combo deals or upsell alongside beverages to increase profitability.

3.3 Strategic Recommendations:

Based on the analysis, the following strategic actions are recommended:

3.3(a) Optimize Pricing & Profitability:

- Increase the price of bread and snacks slightly to enhance profit margins without affecting demand.
- Introduce premium and healthier variants for higher margins.

3.3(b) Focus on High-Performing Categories:

- Scale up production and promotion of pastries and fresh cream items, as they demonstrate strong revenue potential.
- Expand pastry offerings to capitalize on customer demand.

3.3(c) Discontinue or Revamp Weak Products:

- Consider discontinuing dry cakes if profitability remains low.
- Improve dry cake offerings through new flavors, packaging, or special promotions.

3.3(d) Introduce Seasonal & Promotional Strategies:

- Implement festival-based limited editions to drive seasonal demand.
- Offer loyalty programs or discounts for bulk purchases.

3.3(e) Cost & Inventory Optimization:

- Reduce production costs, especially for rich cream products, by sourcing ingredients efficiently.
- Ensure consistent inventory for high-selling products to avoid stockouts.

3.3(f) Conclusion:

The analysis highlights key insights into Golden Crust Bakery's product performance, revealing areas for revenue growth and profitability enhancement. By optimizing pricing, focusing on best-selling items, and making strategic product decisions, the bakery can improve overall financial performance and customer satisfaction.

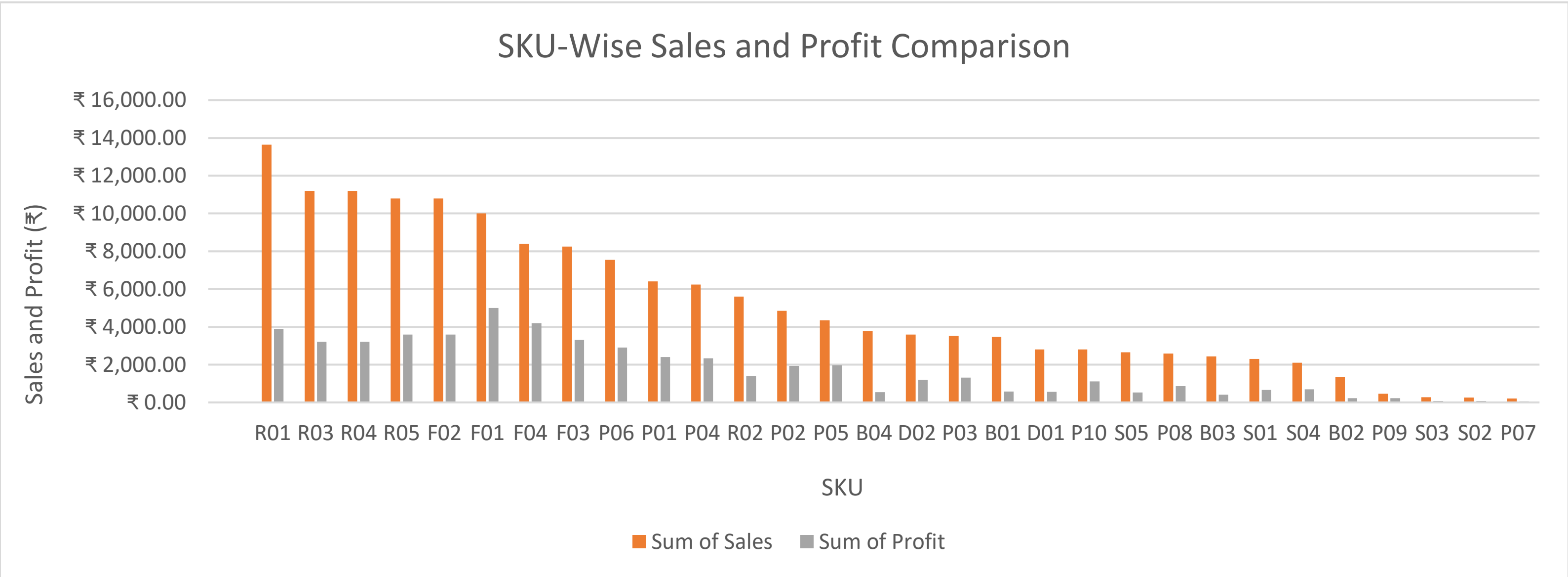

Figure 3(SKU-Wise Sales and Profit Comparison)

3.4 ABC Analysis of Bakery Inventory

3.4(a) Inventory Classification:

- Class A (Top 70% of Sales): Critical products that must always be in stock (e.g., R01, R03, R04, R05, F02, F01, F04, F03).
- Class B (Next 20% of Sales): Moderate stock levels with periodic review (e.g., P02, P05, B04, D02).
- Class C (Final 10% of Sales): Low-priority items to be minimized or discontinued (e.g., S05, P08, B03, S01).

3.4(b) Revenue vs. Profitability Insights:

- High Sales, Low Profit: Items like R01, R03, R04, and R05 generate high sales but low margins, requiring cost optimization or price adjustments.
- Balanced Performance: Products like F01, F04, and F02 have both high sales and strong profitability.
- Low Sales & Low Profit: Items such as S03, S02, and P07 indicate low demand and may need marketing improvements or discontinuation.

3.4(c) Optimization Strategies:

- Cost Optimization: Negotiate better supplier rates or reduce production costs for high-sales, low-margin products.
- Bundling & Promotions: Pair slow-moving products with bestsellers to boost sales and optimize inventory.
- Targeted Marketing: Improve visibility for low-performing items through promotions, discounts, or seasonal offers.
- Discontinuation: Remove consistently underperforming items to save space and reduce costs.

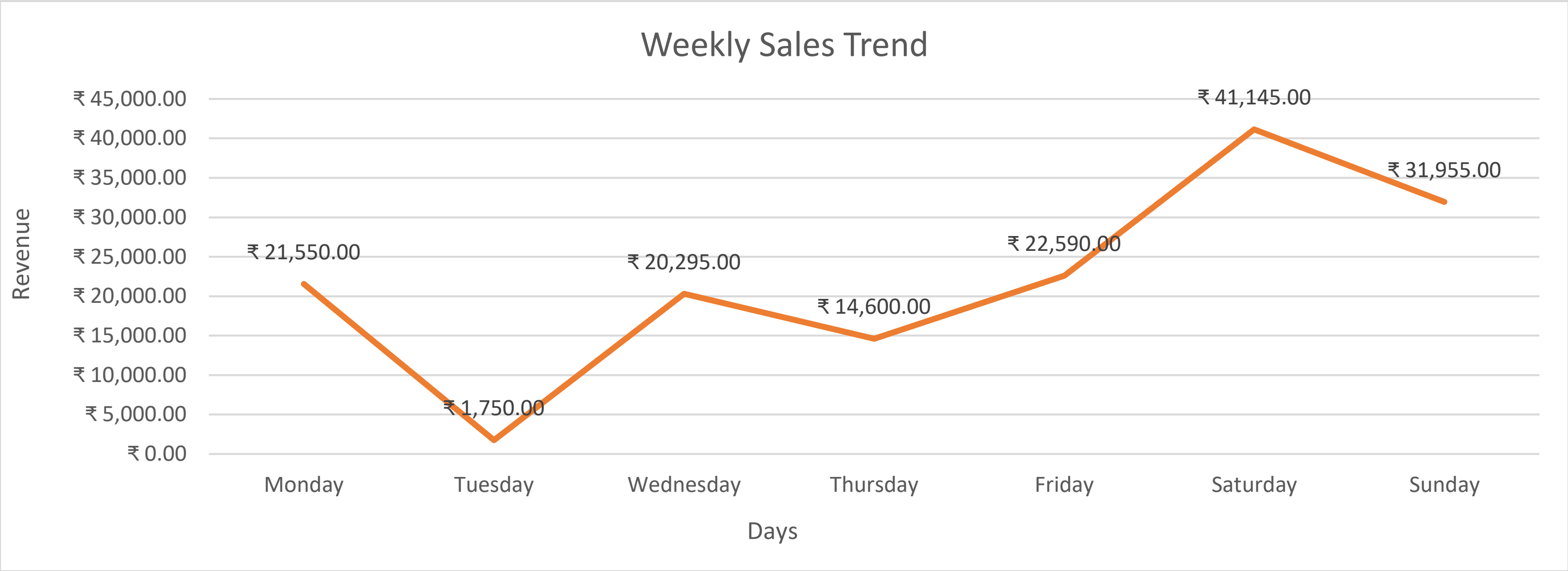

Figure 4(Weekly Sales Trend)

3.5 Business Insights from Weekly Sales Performance:

3.5(a) Peak Sales Occur on Saturdays:

- Saturday records the highest revenue (₹41,145), indicating maximum customer activity and demand.
- Introduce premium pricing, bundle deals, or limited-time weekend specials to maximize revenue. Ensure adequate staffing and inventory to handle the rush efficiently.

3.5(b) Strong Sales on Sundays, but Lower Than Saturdays:

- While Sunday sales (₹31,955) remain high, they are slightly lower than Saturday's peak.
- Consider Sunday-exclusive discounts or morning-time happy hour offers to sustain momentum. Focus on quick service or takeaway deals to match customer shopping patterns.

3.5(c) Sales Drop on Thursdays After a Midweek Recovery:

- Sales recover after Tuesday's off-day, peaking at ₹20,295 on Wednesday but declining to ₹14,600 on Thursday.
- Run midweek special offers on Thursday, such as loyalty discounts or combo meals, to maintain consistency.

3.5(d) Friday Sales Start Picking Up for the Weekend:

- Sales increase to ₹22,590 on Friday, showing early signs of weekend shopping behaviour.
- Introduce pre-weekend promotions, such as "Friday Flash Deals" or discounts on bulk purchases, to further drive sales.

3.5(e) Monday Sales Are Strong, Indicating Post-Weekend Demand:

- Monday sees a healthy ₹21,550 in sales, suggesting customers restock after the weekend.
- Optimize inventory and consider Monday-exclusive meal combos or discounts to attract repeat customers.

3.5(f) Tuesday Closure Provides an Opportunity for Business Optimization:

- With Tuesday being a non-operational day, it serves as a natural break in the business cycle.
- Use Tuesday for restocking inventory, deep cleaning, and staff training to ensure smooth operations for the rest of the week. Also, leverage this time for marketing promotions to announce upcoming deals.

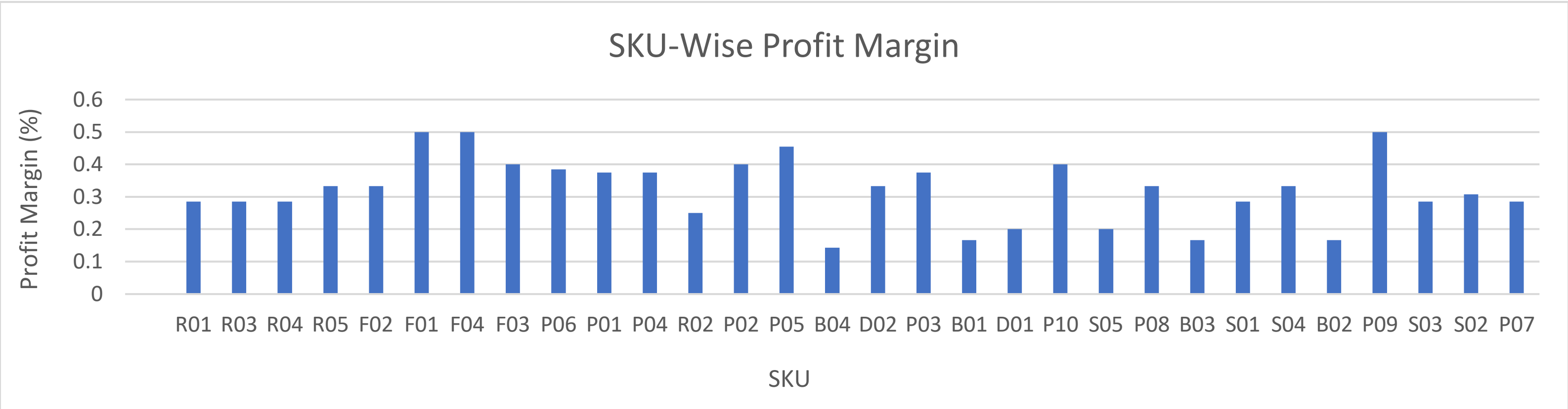

Figure 5 (SKU-Wise Profit Margin)

3.6 SKU-Wise Profit Margin Analysis:

3.6(a) High Profit SKUs (Profit Margin > 40%):

- SKUs: F01, F04, P05, P09
- Insight: These SKUs demonstrate significantly higher profit margins, indicating strong market performance or efficient cost management.

- Recommendation: Invest in strategies to maintain or increase sales volume for these high-profit SKUs, potentially through targeted marketing campaigns or promotions.

3.6(b) Moderate Profit SKUs (Profit Margin 20% - 40%):

- SKUs: R04, R05, P04, R02, P02, B04, D02, P03, D01, P10, S05, P08, S01, S04, B02, S03, S02, P07
- Insight: These SKUs represent a stable profit margin, but may have room for improvement.
- Recommendation: Analyze the pricing and cost structures of these SKUs to identify opportunities for margin enhancement. Consider bundling or cross-selling strategies to boost sales.

3.6(c) Low Profit SKUs (Profit Margin < 20%):

- SKUs: B04, B03, B02
- Insight: These SKUs exhibit low profit margins, potentially due to high costs, low pricing, or low sales volume.
- Recommendation: Conduct a thorough analysis to understand the reasons for low profitability. Consider discontinuing or re-evaluating these SKUs if they consistently underperform or try bundling them with high selling SKUs.

3.6(d) Overall Recommendations:

- Data Integration: Combine profit margin data with sales volume data to gain a more comprehensive understanding of SKU profitability.
- Regular Monitoring: Implement a system for regular monitoring of SKU-wise profit margins to identify trends and make data-driven decisions.

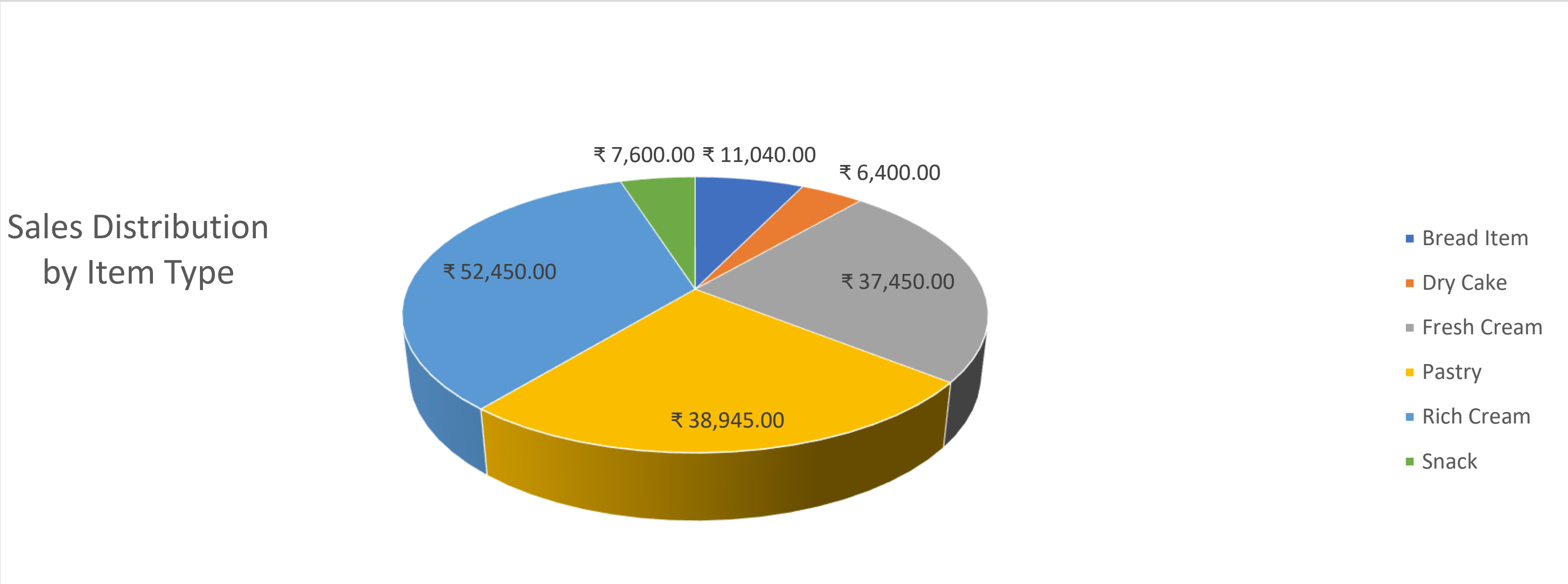

Figure 6(Sales Distribution by Item Type)

3.7 Sales Performance Analysis by Item Type:

3.7(a) Rich Cream Lead Sales Volume:

- Rich Cream (₹52,450) contribute the highest sales, indicating strong consumer demand and frequent purchases.
- Recommendation: Maintain consistent supply, introduce product variations, and explore bulk sales options for increased revenue.

3.7(b) Pastry and Fresh Cream Generate Significant Sales:

- Pastry (₹38,945) and Fresh Cream (₹37,450) hold substantial shares in total sales, proving their popularity.
- Recommendation: Capitalize on this trend by introducing premium flavours, festive offers, and bundling strategies.

3.7(c) Bread Item Sales Are Moderate:

- Bread Item (₹11,040) has a decent sales contribution but is significantly lower than Pastry and Bread Items.
- Recommendation: Improve marketing, introduce limited-time offers, and assess pricing strategies to boost demand.

3.7(d) Dry Cake and Snacks Have the Lowest Sales

- Dry Cake (₹6,400) and Snack (₹7,600) report the lowest sales figures, suggesting limited demand or lack of promotional efforts.

- Recommendation: Reassess customer preferences, explore rebranding, or introduce combo deals to stimulate interest.

3.7(e) Product Strategy Should Focus on High-Selling Items

- The top-selling items (Fresh Cream, Pastry, and Rich Cream) dominate the sales distribution, while lower-selling items need optimization.
- Recommendation: Align inventory, marketing, and production efforts toward top-performing categories while optimizing or repositioning underperforming ones.

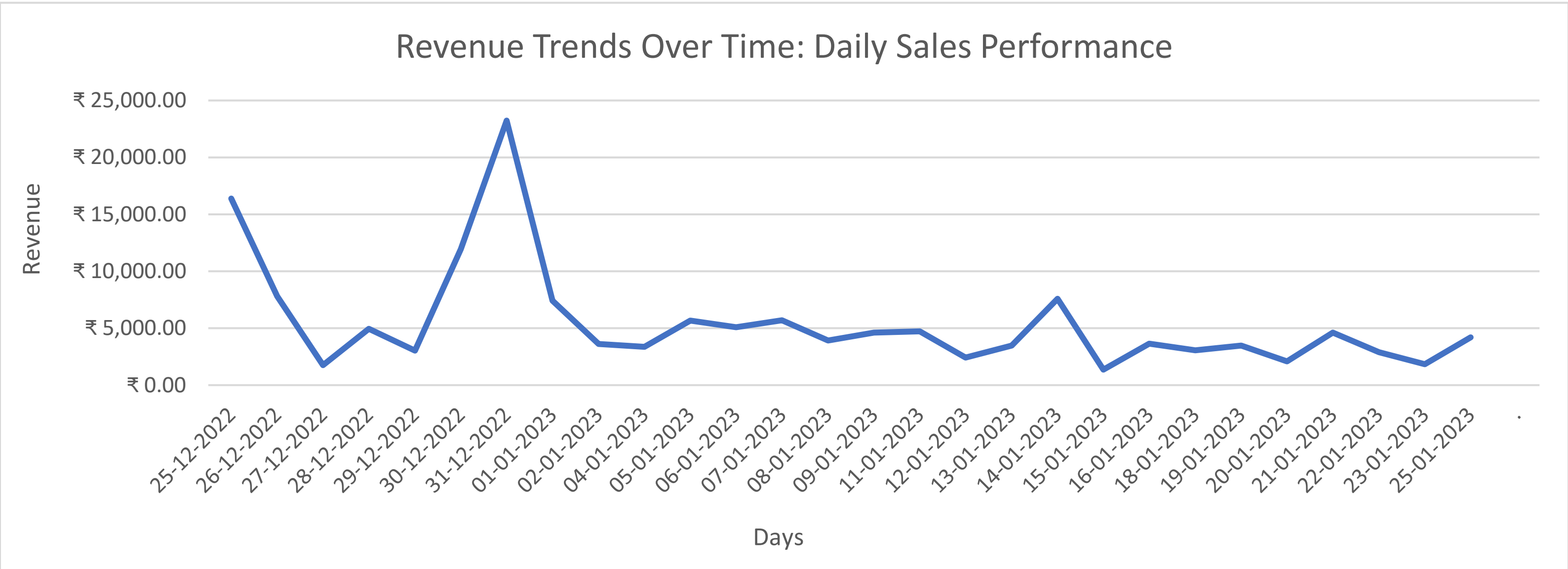

Figure 7 (Revenue Trend)

3.8 Revenue Trends Over Time: Daily Sales Performance

3.8(a) Sales Peak Around Year-End Festivities:

- The graph shows a significant spike in revenue on December 30th and 31st, 2022, likely driven by increased consumer demand due to New Year celebrations. This suggests that seasonal events have a strong influence on sales.

3.8(b) Post-Holiday Stabilization:

- After the New Year peak, sales declined and stabilized at a relatively consistent level. This pattern indicates that while holiday seasons create short-term surges, regular demand remains steady throughout January.

3.8(c) Periodic Fluctuations in January:

- There are minor fluctuations in sales across January, with occasional small spikes. These variations could be attributed to factors such as weekend shopping trends, promotions, or payday effects.

3.8(d) Strategic Recommendations for Bakery Sales Optimization:

Based on the revenue trends, here are five key recommendations to enhance sales and efficiency:

- Capitalize on Peak Demand: Since sales peaked on December 30-31st, introduce New Year-exclusive product bundles and offer pre-order discounts to boost advance sales.
- Optimize Staffing & Inventory: Ensure adequate stock of high-demand items and schedule extra staff during peak periods to handle increased orders smoothly.
- Extend the Festive Sales Window: Offer post-New Year promotions, such as limited-time discounts or loyalty rewards, to sustain momentum into January.
- Targeted Marketing Campaigns: Use social media and email marketing to promote bestsellers, emphasizing limited-edition products and early-bird offers.
- Encourage Sales During Off-Peak Days: Introduce weekday specials or bulk-order discounts to drive sales when demand is lower.

4 Interpretation of Results and Recommendations:

4.1 Key Findings & Interpretations:

4.1(a) Revenue & Profit Trends:

- Rich Cream (₹52,450 revenue): High sales but moderate margins (₹15,300 profit).
    - Interpretation: High demand justifies inventory prioritization but requires cost optimization.
- Fresh Cream (₹37,450 revenue): Best margins (₹16,100 profit).
    - Interpretation: Ideal for bundling and premium pricing.

- Bread & Dry Cake: Minimal contribution (₹11,040 and ₹6,400 revenue).
    - Interpretation: Consider optimizing costs or exploring new vendors for better pricing.

4.1(b) ABC Inventory Classification:

- Class A (R01, F02): Contributed 70% of sales.
    - Interpretation: Critical to avoid stockouts; maintain buffer stock.
- Class C (S05, B03): 10% sales contribution.
    - Interpretation: Minimize stock or phase out.

4.1(c) Weekly Sales Trends:

- Peak Days: Saturday (₹41,145) and Sunday (₹31,955).
    - Interpretation: Introduce weekend-specific bundles and dynamic pricing.
- Midweek Dip (Thursday: ₹14,600):
    - Interpretation: Stimulate demand with loyalty discounts.

4.1(d) High-Volume, Low-Profit SKUs:

- Examples: R01, P01 (160 units sold but low margins).
    - Interpretation: Leverage volume to drive sales of high-margin items via bundling.

4.2 Strategic Recommendations:

4.2(a) Product & Pricing Strategy:

- Prioritize High-Margin SKUs:
    - Promote Fresh Cream (F01, F04) through “Premium Fridays” and bundling (e.g., “Coffee + Fresh Cream Pastry”).
- Optimize Low-Margin, High-Volume Items:
    - Renegotiate supplier costs for Rich Cream SKUs (R01, R03).
    - Test a 5–7% price increase for high-demand, low-margin items.

4.2(b) Inventory Management:

- ABC-Driven Stocking:
    - Class A: 7-day buffer stock with real-time tracking.
    - Class C: Drop-shipping or custom orders only.
- Just in Time Ordering: Reduce waste for perishables (e.g., Fresh Cream).

4.2(c) Promotional Campaigns:

- Weekend Specials:
    - “Saturday Mega Bundles” (Rich Cream + Pastry at 15% off).
    - “Sunday Morning Happy Hour” (20% off pre-10 AM orders).
- Midweek Recovery:
    - “Thursday Loyalty Discounts” (10% off for repeat customers).

4.2(d) Operational Efficiency:

- Staffing Adjustments:
    - Increase weekend workforce by 30% and cross-train staff.
    - Use Tuesdays (non-operational) for inventory audits and training.
- Dynamic Pricing:
    - Add a 5% premium on Rich Cream during weekend rushes.

4.2(e) Customer Retention:

- Loyalty Programs:
    - “Buy 5 Rich Creams, Get 1 Free” to encourage repeat purchases.
- Feedback Integration:
    - QR code surveys at checkout to refine product offering